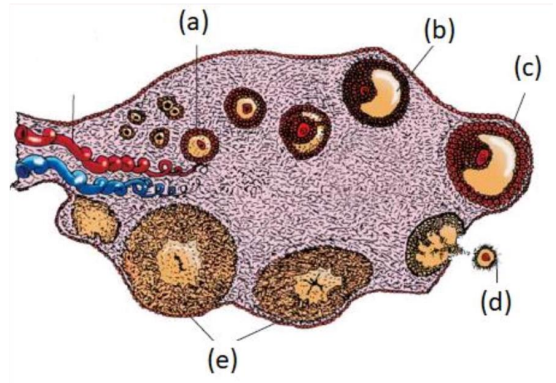


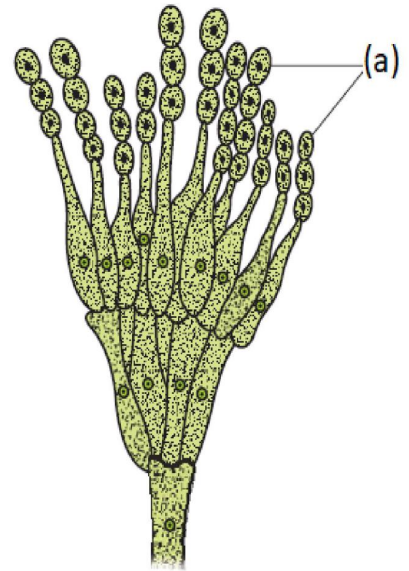
I. Mark the option with correct label

- a. Secondary follicle
- b. Tertiary follicle
- c. Corpus luteum
- d. Graafian follicle



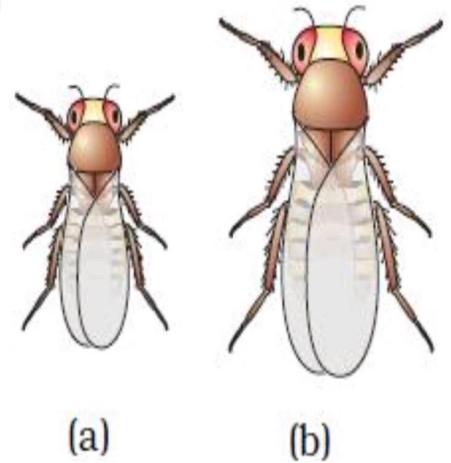
2. What does the marked structure represent

- a. Budding
- b. Conidia
- c. Spores
- d. Gemmules



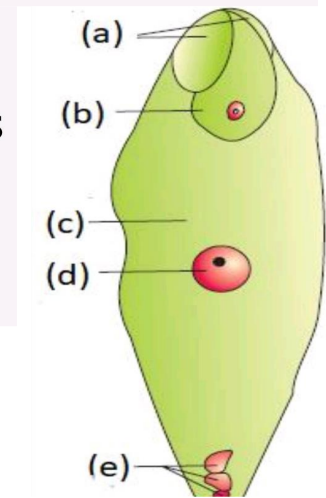
3. The diagram shows the fruit flies used by Morgan for his experiment. Mark the correct statement.

- a. They complete their life cycle in 2 months, thus many life cycles could be studied.
- b. Scientific name is *Drosophila melanogaster*.
- c. Male and female are easily distinguishable.
- d. Single mating produces single offspring.



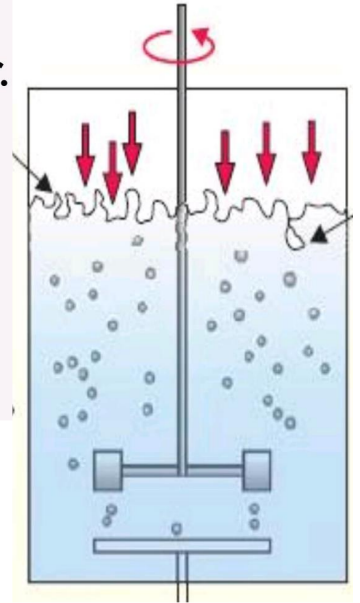
4. The diagram is of a fertilized embryo sac. Mark the correct statement.

1. (a) represents the antipodal cells
2. (c) represents Primary Endosperm Nucleus
3. (b) represents egg cell
4. (e) represents degenerating antipodal cells



5. Mark the incorrect statement

- a. Diagram represents simple stirred tank reactor.
- b. Oxygen is available throughout the reactor.
- c. Bubble increases the surface area for oxygen transfer.
- d. Sterile air is used in this reactor.



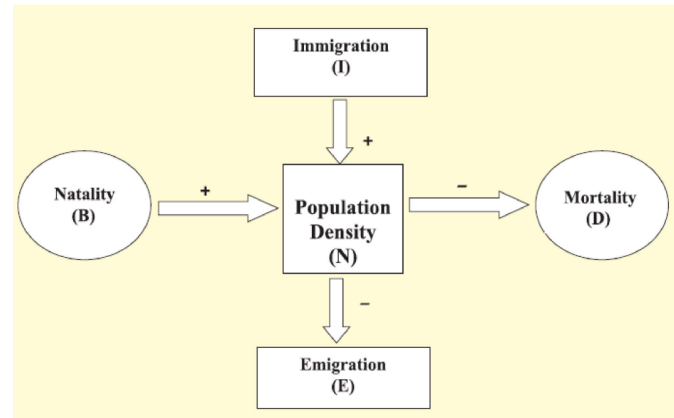
6. If population density at time t is n , the its density at time $t+1$ (represented by N) is

a. $N = n + [(B+I) + (D+E)]$

b. $N = n + [(B+E) - (D+I)]$

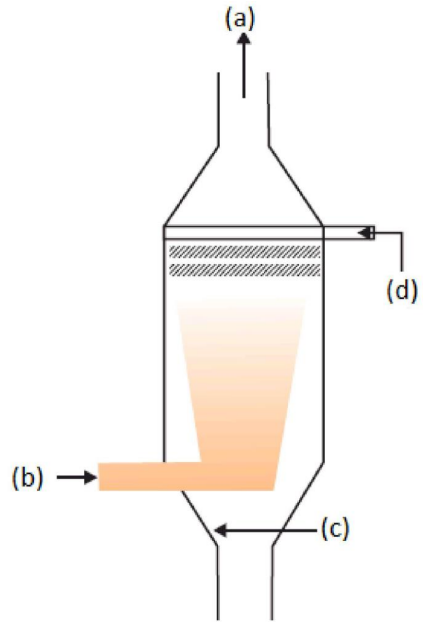
c. $N = n + [(D+I) - (B+E)]$

d. $N = n + [(B+I) - (D+E)]$



7. Mark the incorrect marking

- a. Clean air
- b. Dirty air
- c. Particulate matter
- d. Charged plate



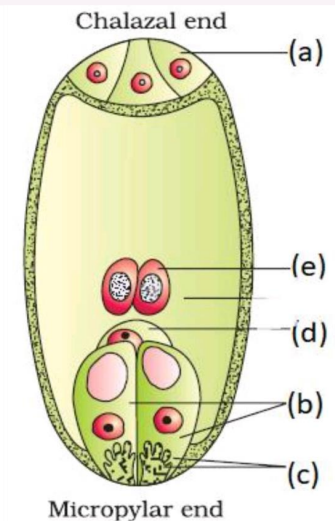
8. Which of the following structure guides the pollen tube into the mature embryo sac

1. (a)

2. (b)

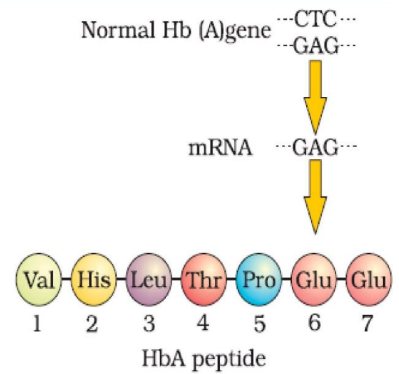
3. (c)

4. (d)



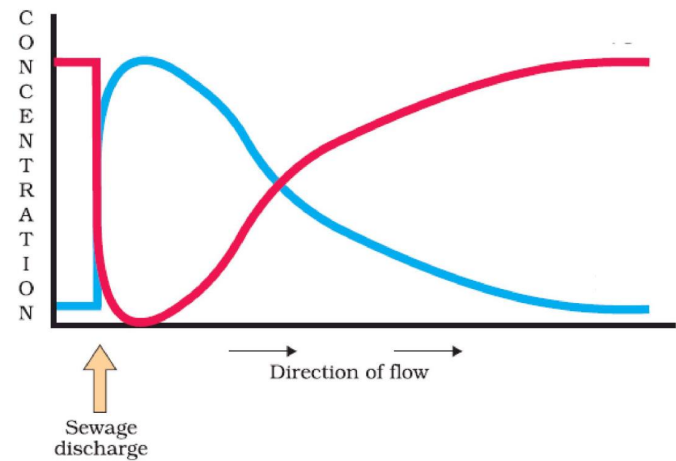
9. Mark the incorrect statement about the diagram.

- a. It represents normal HbA peptide
- b. Sick cell anemia occurs if Glu at 6 is replaced by Valine
- c. The codon coding for glutamic acid is GAG
- d. In sickle cell anemia GAG of mRNA gets converted to GTG



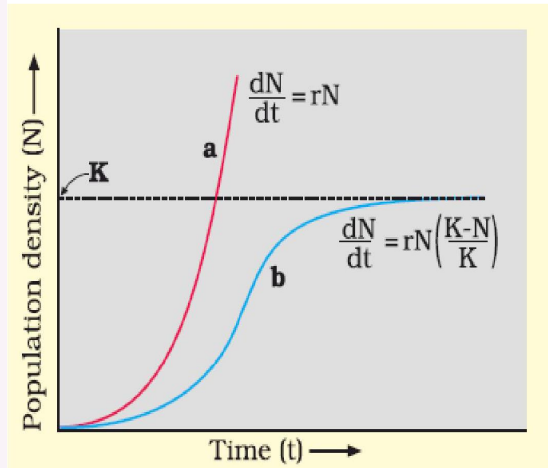
10. What does the Blue line represent

- a. Dissolved oxygen
- b. BOD
- c. Biomagnification
- d. Pollution level



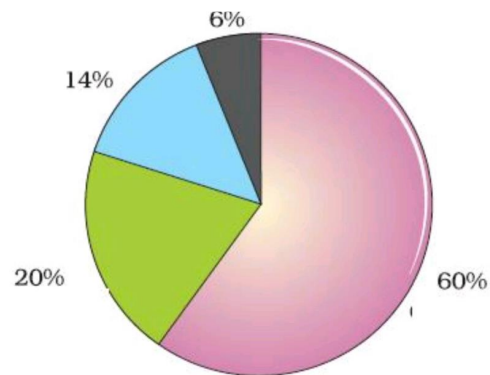
11. The diagram represents a population growth curve. Mark the correct option.

1. (a) represents the curve when responses are limiting
2. (b) represents the curve when the responses are not limiting
3. “r” is the intrinsic rate of natural decrease
4. “k” is the carrying capacity



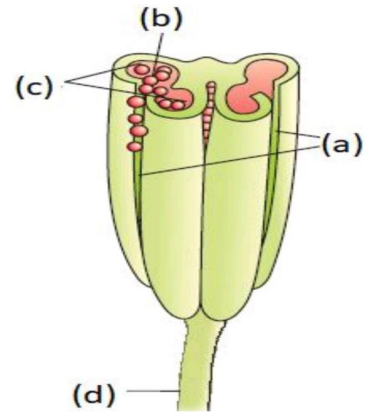
12. The diagram shows the relative contribution of various gases to global warming. Mark the correct composition of gases

- a. Methane – 14%
- b. CFCs – 20%
- c. CO₂ – 60%
- d. N₂O – 6%



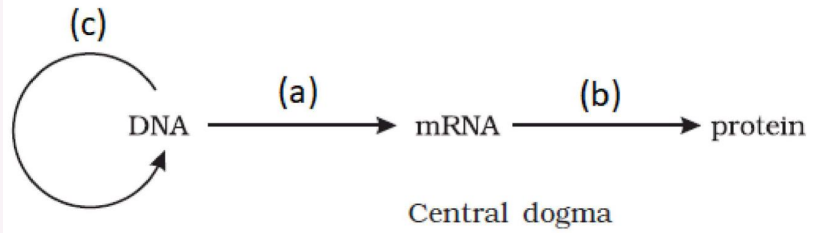
13. What is the incorrect about (b).

- a. When mature contains 2 cells.
- b. Develops from microspore.
- c. While developing, it derives nutrition from
- d. None of these



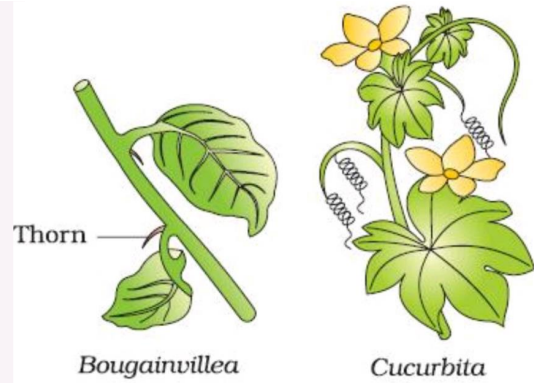
14. An exception to the above dogma is

- a. Retrovirus
- b. Rhinovirus
- c. Arbovirus
- d. Rabdovirus



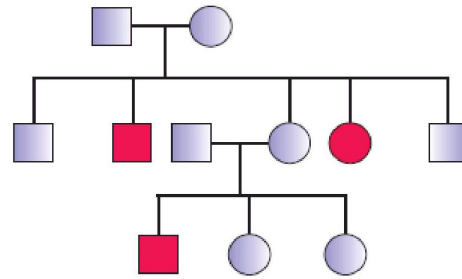
15. The diagram represents

- a. Homologous evolution
- b. Analogous evolution
- c. Both a and b
- d. None of these



16. The pedigree chart represents-

- a. Autosomal dominant
- b. Autosomal recessive
- c. Y linked trait
- d. X linked dominant

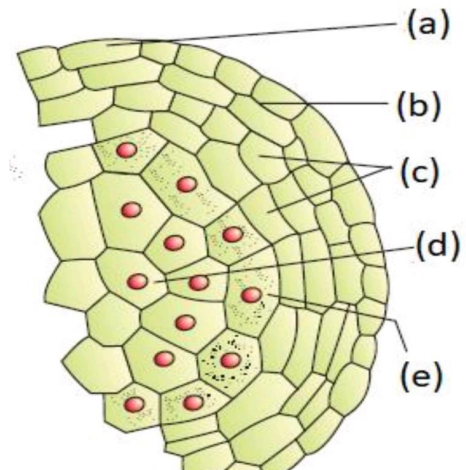


(b)



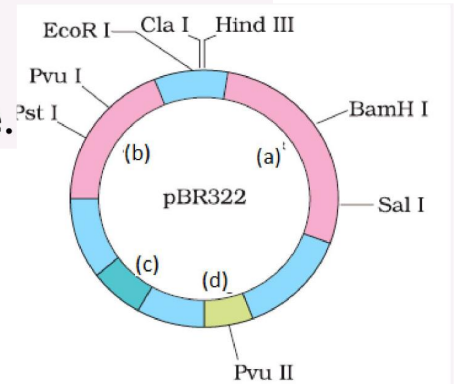
17. Which layer serves the function of providing nutrition to the developing pollen

1. (c)
2. (b)
3. (e)
4. (d)



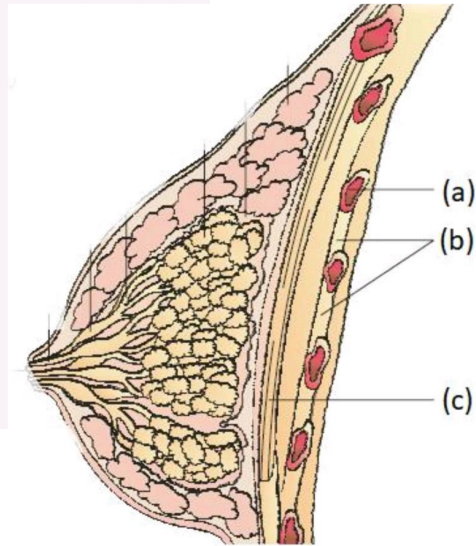
18. Mark the incorrect statement

1. pBR322 is a cloning vector of E.coli
2. (a) represents the tetracycline resistance gene
3. (c) represents rop
4. (b) represents the ampicillin resistance gene.



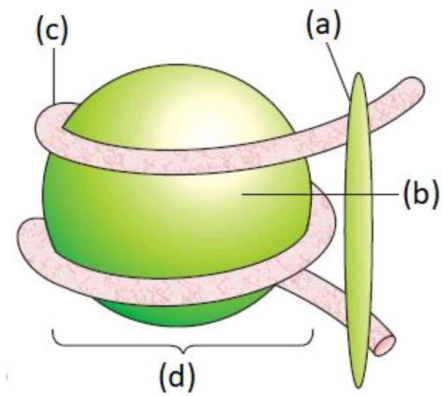
19. Which the muscle marked by (c)

- a. Latismus
- b. Pectoralis minor
- c. Pectoralis major
- d. Trapezius



20. The diagram represents a nucleosome. Mark the correct labelling

- a. HI Histone
- b. Core of histone
- c. Positively charged DNA
- d. Histone octamer



21. Mark the correct answer

a. Competition

b. Mutualism

c. Predation

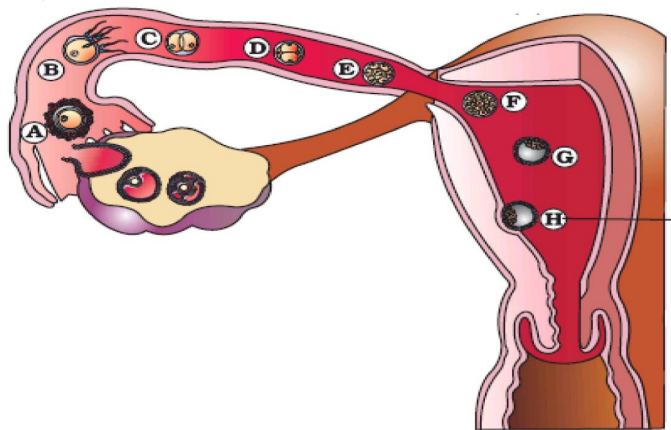
d. Commensalism

Species A	Species B	Name of Interaction
+	+	(a)
-	-	(b)
+	-	(c)
+	-	(d)
+	0	(e)
-	0	(f)



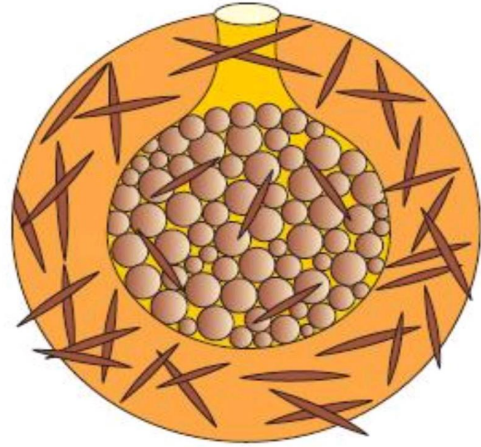
22. What is the location of Morula in the diagram

- a. C
- b. E
- c. G
- d. D



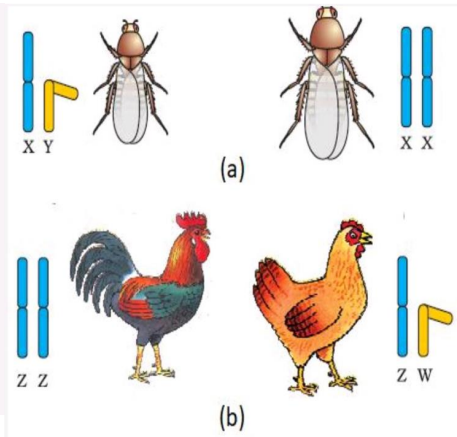
23. Diagram represents

- a. Zoospores
- b. Gemmules
- c. Buds
- d. Spores



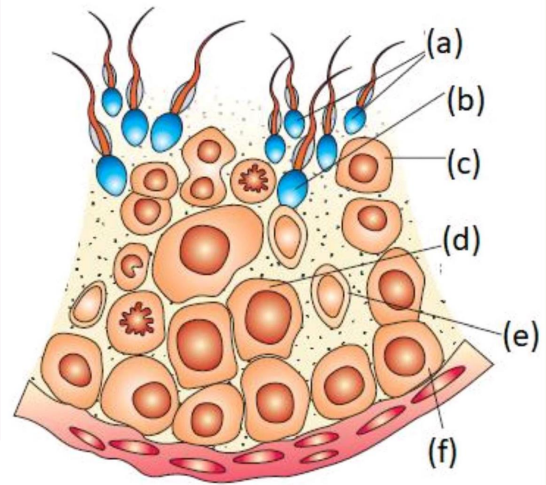
24. Mark the correct statement

- a. Males are homogametic in *Drosophila*
- b. Females are homogametic in birds
- c. Male are homogametic in birds
- d. None of these.



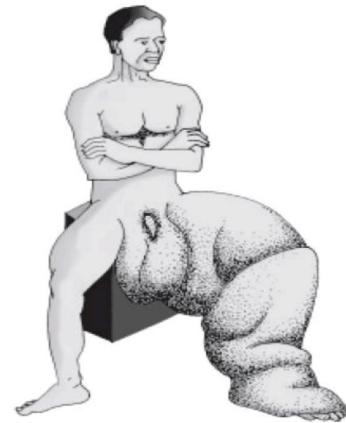
25. Mark the correct label

1. (a)- Spermatid
2. (d)- secondary spermatocytes
3. (f)- spermatogonium
4. (e) primary spermatocyte



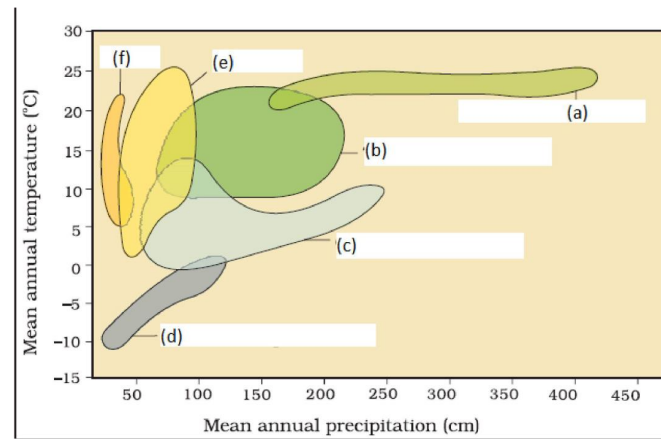
26. What is the vector for the disease shown in the figure

- a. Culex
- b. Anopheles
- c. Aedes
- d. Sandfly



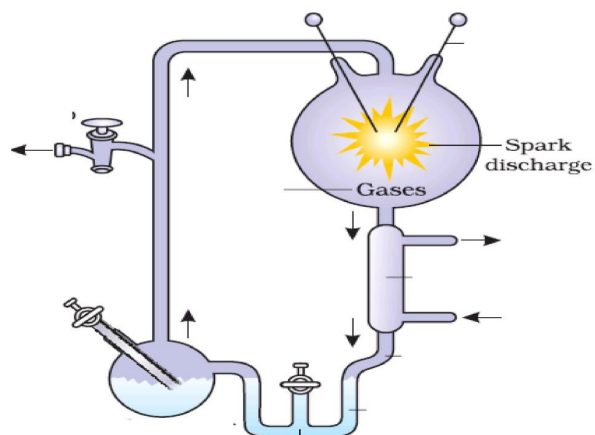
27. The diagram represents biome distribution according to temperature and precipitation. Mark the correct option

1. (a) Coniferous forest
2. (b) Temperate forest
3. (f) Alpine tundra
4. (d) Desert



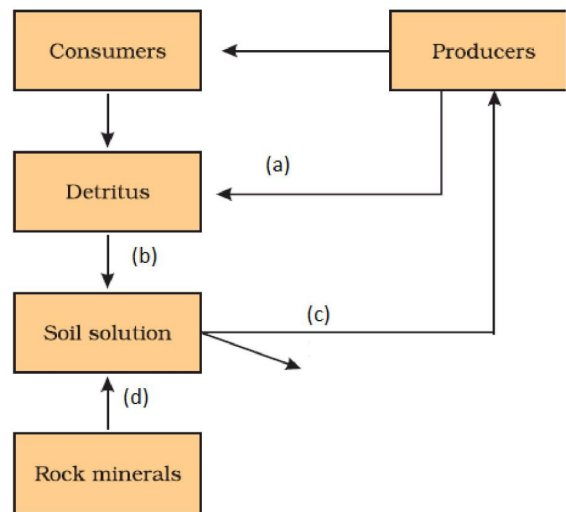
28. Which of the following were the gases involved in this experiment

- a. CH_4
- b. NH_3
- c. H_2O
- d. All the above



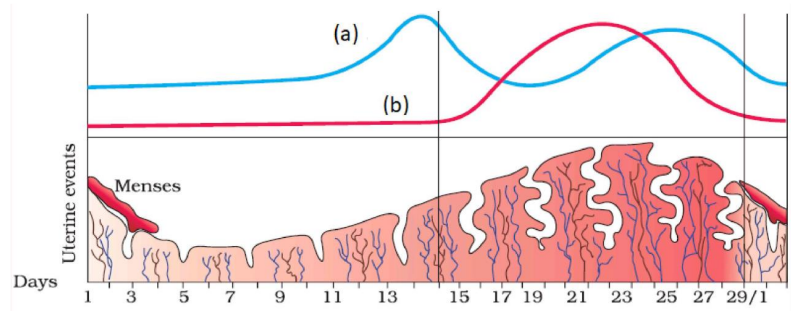
29. Mark the correct option

- a. Decomposition
- b. Litter fall
- c. Run off
- d. Weathering



30. Which hormone is represented by (a)

1. LH
2. FSH
3. Estrogen
4. Progesterone



Answer Key

- 1.b
- 2.b
- 3.c
- 4.d (4)
- 5.a
- 6.d
- 7.d
- 8.c (c)
- 9.d
- 10.b

Answer Key

- 11.d (4)
- 12.d
- 13.d
- 14.a
- 15.a
- 16.b
- 17.c (3)
- 18.c (3)
- 19.c
- 20.a

Answer Key

- 21.c
- 22.b
- 23.b
- 24.c
- 25.c (3)
- 26.a
- 27.b (2)
- 28.d
- 29.d
- 30.c (3)